

PROFILE

Jinmin Kim

Robot Software Engineer - 3D Registration, Medical Imaging, Robot Systems

An R&D engineer whose work spans surgical navigation, XR, 3D registration, multi-sensor integration, and autonomous-forklift field deployment. I connect coordinate models and algorithms to physical devices, user interfaces, and explicit validation flows.

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EXPERIENCE AND EDUCATION

2023-02 - Present	DIGITRACK Inc.	Researcher · Robot Software Surgical-navigation software with Samsung Medical Center; NeuroPilot rTMS; SKADI API and Viewer; DOTORI vision and safety policy; respiratory tracking (2026-).
2021-03 - 2023-02	DGIST	M.S. in Robotics Occlusion-based mandibular fracture reduction planning (JIIM 2024, joint first author) and AR surgical-navigation and XR research.
2020-07	UNIST U-SURF	Summer Research Intern Reviewed soft-robotics literature and conducted magnetic-force experiments.
2015-03 - 2021-02	Kumoh National Institute of Technology	B.S., Mechanical System Engineering System-vision undergraduate research, a quadruped robot design-and-build, and invention and patent activity.

USAGE-BASED ENGINEERING EXPERIENCE

3D Registration and Optimization Feature, surface, and coordinate-frame registration plus sensor-precision validation with PCA, ICP, CLPSO, Open3D, OpenCV, and SciPy.	Medical Navigation and XR Working navigation and XR applications with 3D Slicer, VTK, Qt, OpenIGTLink, Unity, MRTK, Meta SDK, and Photon.
Robot and Sensor Integration Integrated ROS 2, Zenoh, Isaac Sim, NDI tracking, HoloLens 2, Meta Quest, RGB, ToF, and LiDAR, and deployed them in the field.	Application-layer Engineering Algorithms, API and Viewer software, XR apps, ROS 2 packages, and SAM-family segmentation PoCs in Python, C#, and C++.

Research and Public Evidence

2024 / JOINT FIRST AUTHOR

A Proof of Concept: Optimized Jawbone-Reduction Model for Mandibular Fracture Surgery

Journal of Imaging Informatics in Medicine, Q1 SCIE

Public article

2022 / INTERNATIONAL CONFERENCE PRESENTATION

Dental Occlusion Model Using Arch Line for Mandibular Fracture Surgery

ACCAS 2022

ACHIEVEMENT SIGNALS

PATENTS 7 applications / 3 grants	AWARDS 9 total	LANGUAGES Korean - Native / Japanese - JLPT N2
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SELECTED AWARDS

- 2024 Best Poster Award, Korean Society of Medical Metaverse
- 2023 Best Paper Award, Korean Society of Medical Robotics
- 2019 Grand Prize, International TRIZ Competition

REPRESENTATIVE WORK SCOPE

Surgical Navigation / Mandibular Fracture Optimization / Life Careverse / rTMS Navigation / Surface-guided Respiratory Tracking / SKADI Software / Autonomous Forklift Registration / AI Build Lab

Two-page CV using only public facts approved on 2026-08-21.